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STEPS TAKEN

1. Project Initialization

- The project was created using the new React Native Community CLI command:
npx @react-native-community/cli init MobileApp.
- This command generated the complete project structure, including the native android and ios folders.

2. Folder Structure and Code Organization

- A src directory was created to house all source code.
- Inside src, the code was further organized into two subdirectories:
 - screens: Contained the UI files for each page (LandingScreen.js, LoginScreen.js, DashboardScreen.js).
 - navigation: Contained the navigation logic in a single file (AppNavigator.js).
- The main App.tsx file was then updated to render the AppNavigator.

3. Dependency Installation

- The necessary libraries for screen navigation were installed using npm:

```
npm install @react-navigation/native @react-navigation/native-stack react-native-screens  
react-native-safe-area-context
```

4. Environment Configuration

- The ANDROID_HOME environment variable was configured on the system to point to the correct Android SDK location.
- The system's Path variable was updated to include the Android SDK's platform-tools and emulator directories, making tools like adb accessible from the command line.

5. Native Code Modification for Android

- To ensure compatibility with the react-native-screens library, a minor change was made to android/app/src/main/java/com/mobileapp/MainActivity.kt by adding the required onCreate method.

CHALLENGES ENCOUNTERED

1. Outdated CLI Conflict

- **Error:** TypeError: cli.init is not a function

- **Cause:** An old, globally installed version of react-native-cli was conflicting with the modern npx command.
- **Solution:** The old global package was uninstalled (npm uninstall -g react-native-cli), and the project was re-initialized using the correct community CLI command.

2. **SDK Location Not Found**

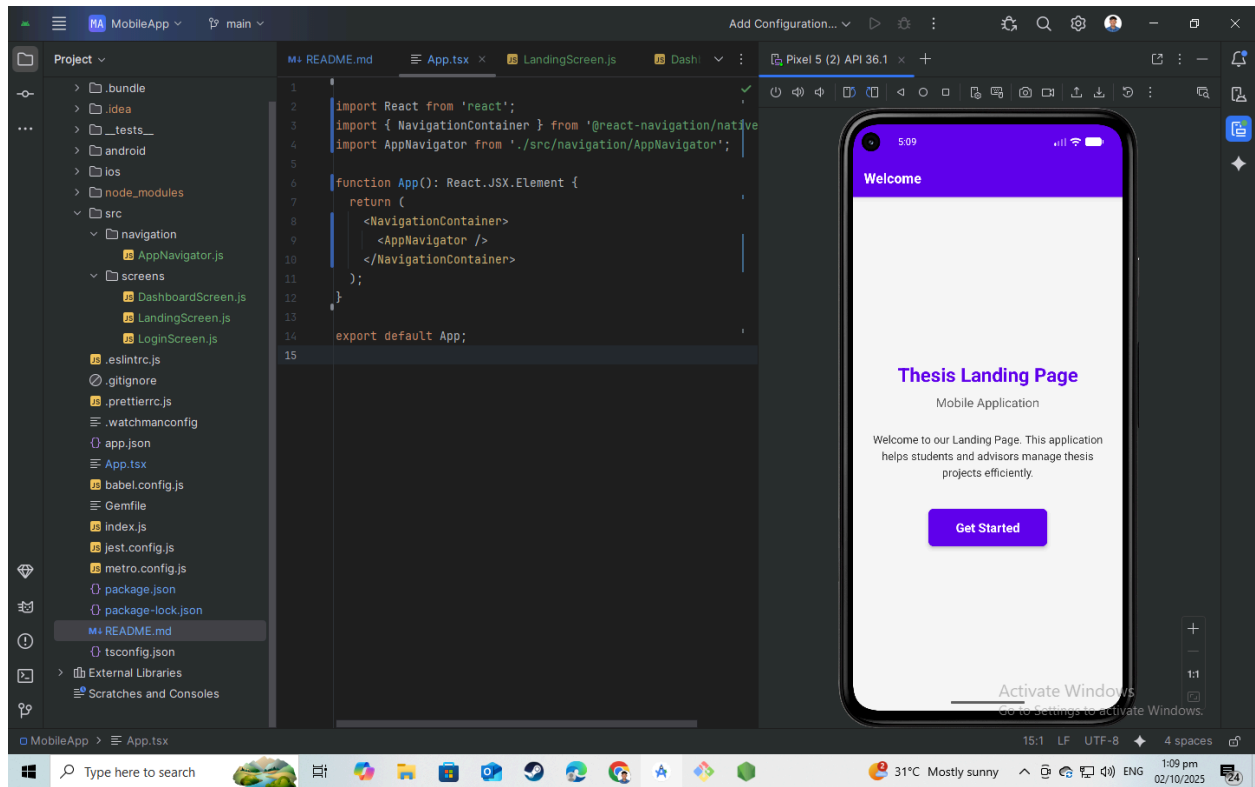
- **Error:** SDK location not found. Define a valid SDK location with an ANDROID_HOME...
- **Cause:** The Gradle build system could not find the Android SDK on the development machine.
- **Solution:** The ANDROID_HOME environment variable was correctly set to the SDK path, which allowed the build to proceed.

3. **Application Crashing on Launch**

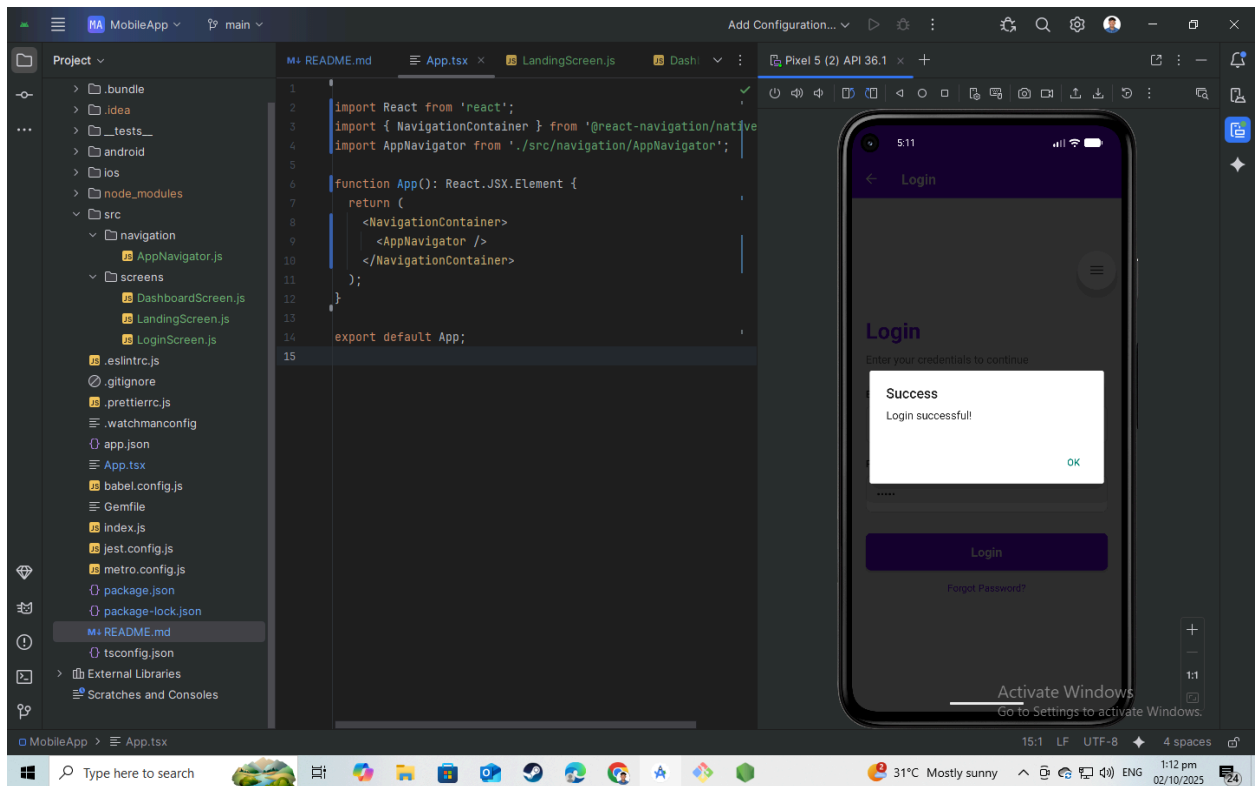
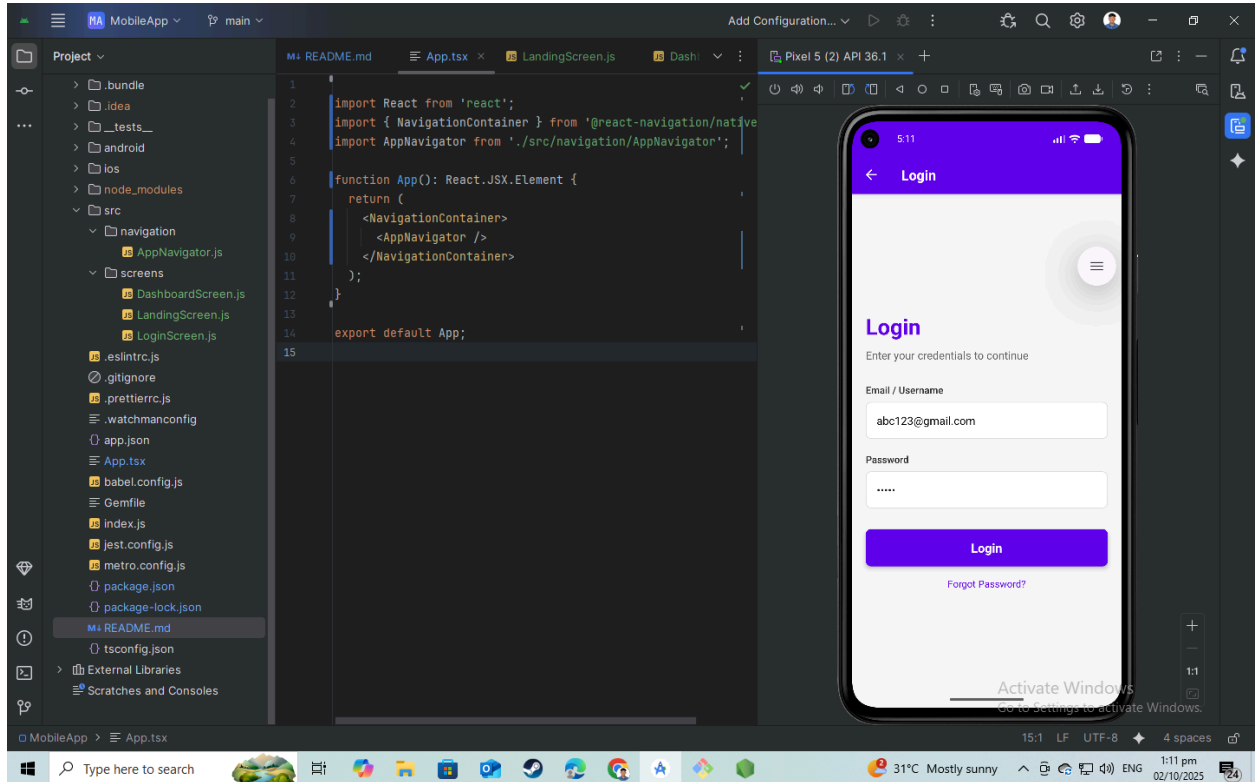
- **Error:** The app built successfully but crashed immediately upon opening on the emulator.
- **Cause:** The react-native-screens library (a dependency for React Navigation) requires native configuration that was missing from the default template.
- **Solution:** The required onCreate method was added to the MainActivity.kt file, which stabilized the application and enabled the navigation to function correctly.

SCREENSHOTS

Landing Page



Login Screen



Dashboard

